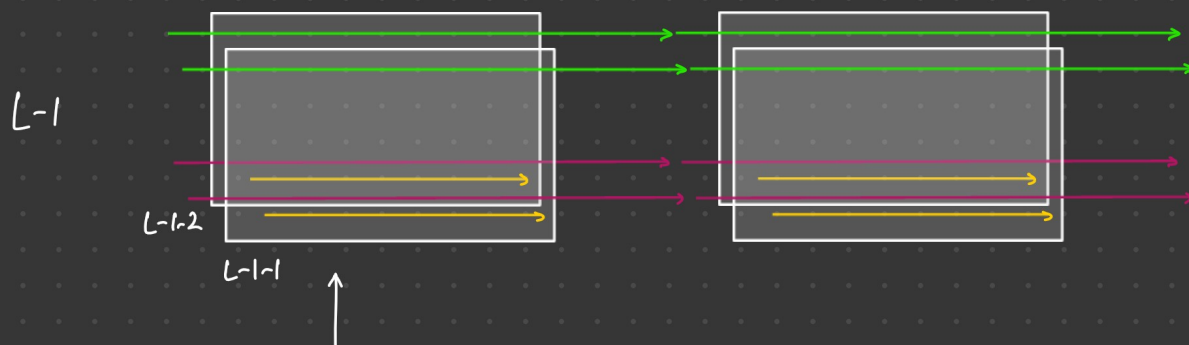


Agenda - 22-02-2025

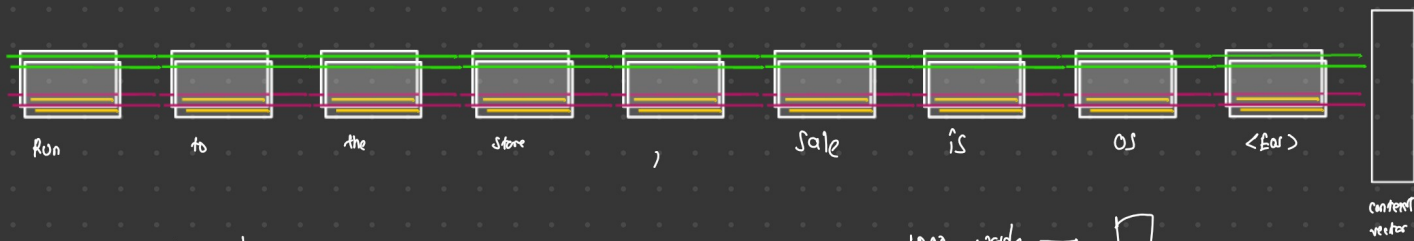
- Attention
- Transformer
 - Single head attention
 - Multi head attention



run to the store, sale is on <EOS>

T → 0 1 2 3 4 5 6 7 8

ATTN → 2
LSTM → 50-100

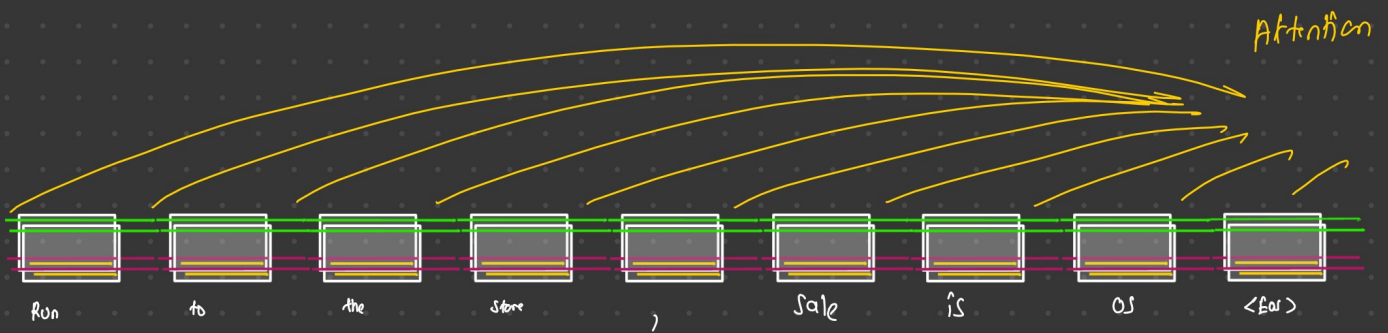
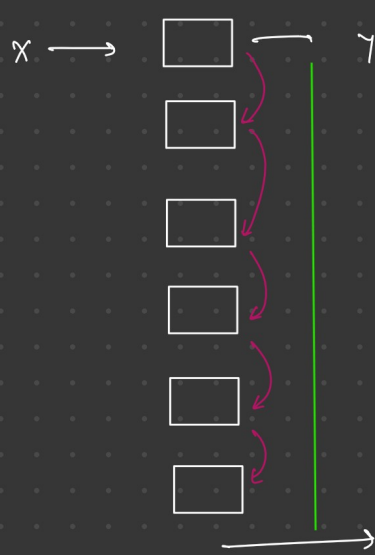


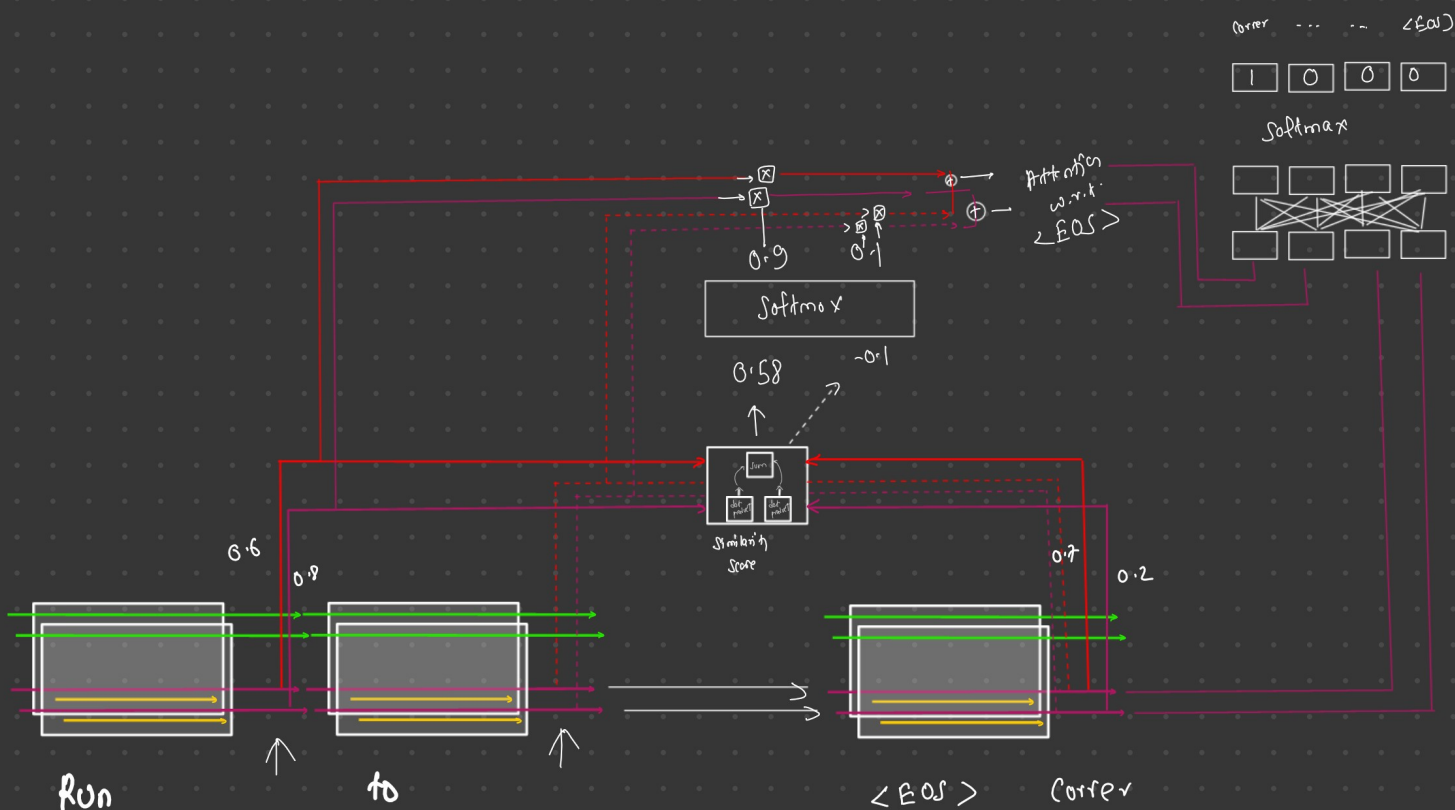
↓ ↓
~~run to~~ The store, sale is on
↑ ↑

→ 1000 words →



→ hidden state - short term memory
 → cell state - long term memory





cosine similarity:

$$\frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Run $\leftarrow A = \begin{bmatrix} A_1 & A_2 \\ 0.6 & 0.8 \end{bmatrix}$

<EOS> $\leftarrow B = \begin{bmatrix} B_1 & B_2 \\ 0.7 & 0.2 \end{bmatrix}$

Calculation:

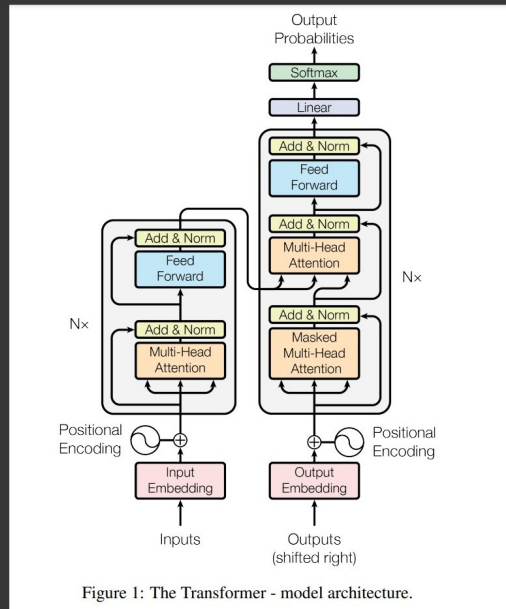
$$(0.6 \times 0.7) + (0.8 \times 0.2) = 0.58$$

Attention $\rightarrow$ sentence w.r.t. your output token

Content vector $\rightarrow$ encoded vector

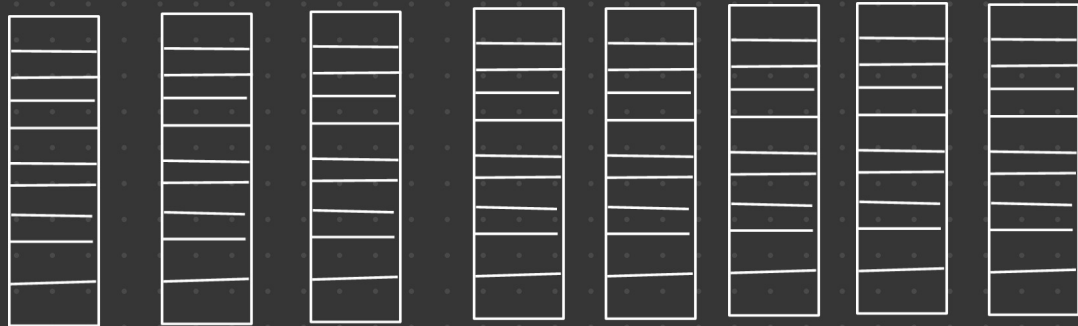
research paper: — Attention is all you need

— Transformer



(1) Calculate Embedding

Run to the store, Sale is on.



10 dim
embedding

$$d_{\text{model}} = 10$$

(2) positional encoding



$$PE(pos, 2i) = \sin \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

$$PE(pos, 2i+1) = \cos \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

pos - position index of the word in the sequence

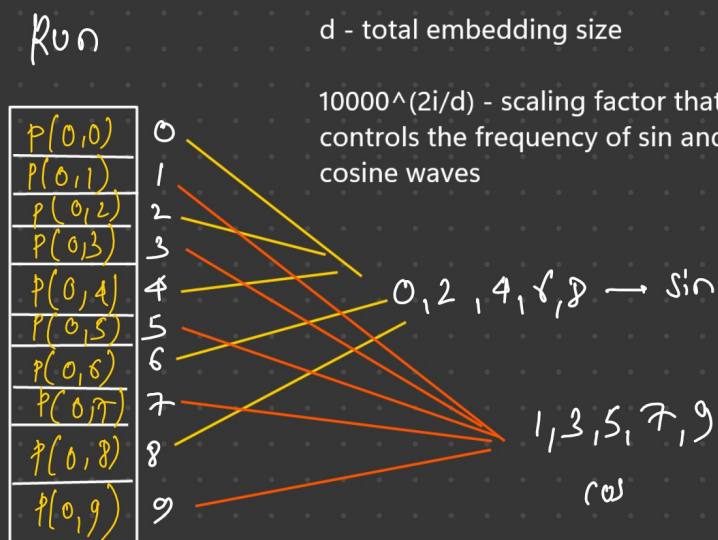
i - index of the embeddings dimension

d - total embedding size

$10000^{(2i/d)}$ - scaling factor that controls the frequency of sin and cosine waves

$i=0$

$$PE(pos, 0) = \sin \frac{0}{10000^{\frac{2 \times 0}{10}}} = \leftarrow$$



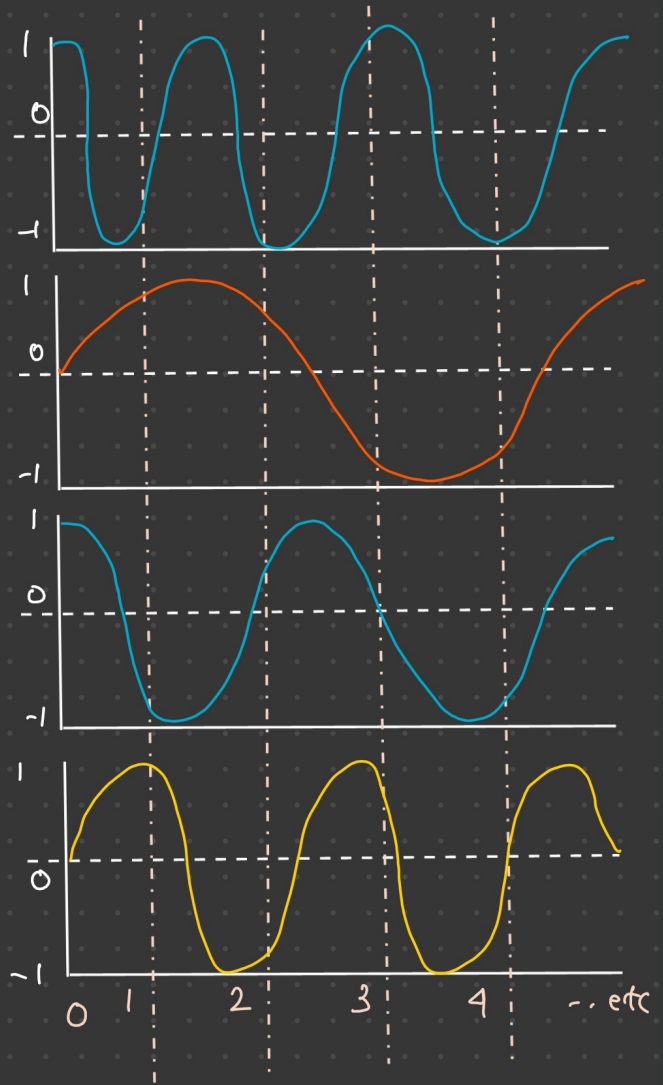
$$\begin{aligned}
 i=0, & \quad 10000^0 = 1 \\
 i=1, & \quad 10000^{0.2} = \times \\
 i=2, & \quad 10000^{0.4} =
 \end{aligned}$$

$\cos 2$

$\sin 2$

$\cos 1$

$\sin 1$



$p(0,0)$	$\xrightarrow{\sin}$	0
$p(0,1)$	$\xrightarrow{\cos}$	-1
$p(0,2)$	$\xrightarrow{\sin}$	0
$p(0,3)$	$\rightarrow$	-1
$p(0,4)$	$\rightarrow$	0
$p(0,5)$	$\rightarrow$	-1
$p(0,6)$	$\rightarrow$	0
$p(0,7)$	$\rightarrow$	-1
$p(0,8)$	$\rightarrow$	0
$p(0,9)$	$\rightarrow$	1

$p(1,0)$	$\xrightarrow{\sin}$	1
$p(1,1)$	$\xrightarrow{\cos}$	-0.9
$p(1,2)$	$\xrightarrow{\sin}$	0.7
$p(1,3)$	$\rightarrow$	-0.6
$p(1,4)$	$\rightarrow$	$\times$
$p(1,5)$	$\rightarrow$	$\times$
$p(1,6)$	$\rightarrow$	$\times$
$p(1,7)$	$\rightarrow$	$\times$
$p(1,8)$	$\rightarrow$	$\times$
$p(1,9)$	$\rightarrow$	$\times$

$\rightarrow$ Embedding is of shape 18

$\sin \rightarrow 5$

$\cos \rightarrow 5$

$$\boxed{512} \rightarrow$$

sim $512/2$

(0) $512/2$

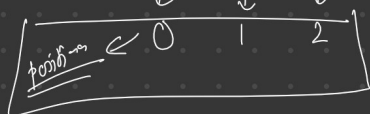
Embedding

positional information

positional embedding

0.9	$P(0,0)$	0.9
0.5	$P(0,1)$	0.5
0.4	$P(0,2)$	0.4
0.2	$P(0,3)$	0.2
0.1	$P(0,4)$	0.1
0.35	$P(0,5)$	0.35
0.7	$P(0,6)$	0.7
0.8	$P(0,7)$	0.8
0.96	$P(0,8)$	0.96
0.22	$P(0,9)$	0.22

sentence — run to store — — —



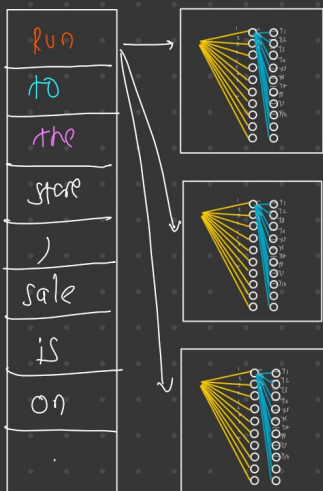
Hi, How are you

position

sim —
(0) —

embedding + position
→

10×10

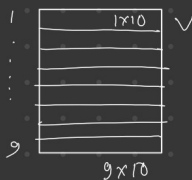
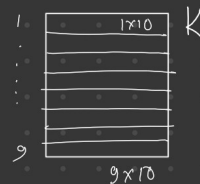
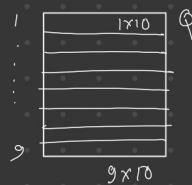


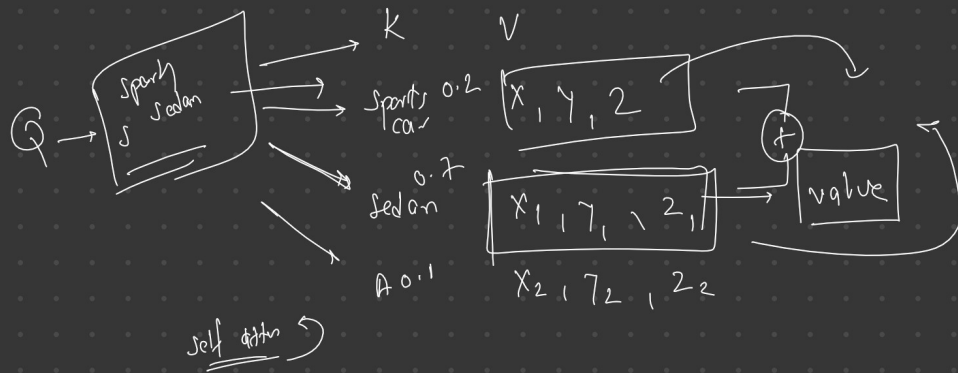
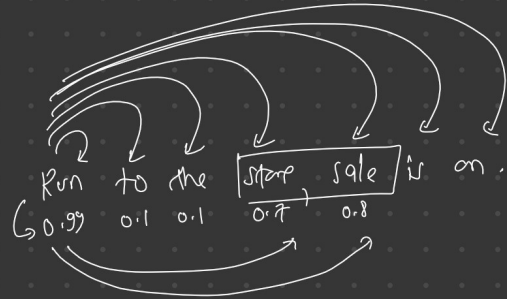
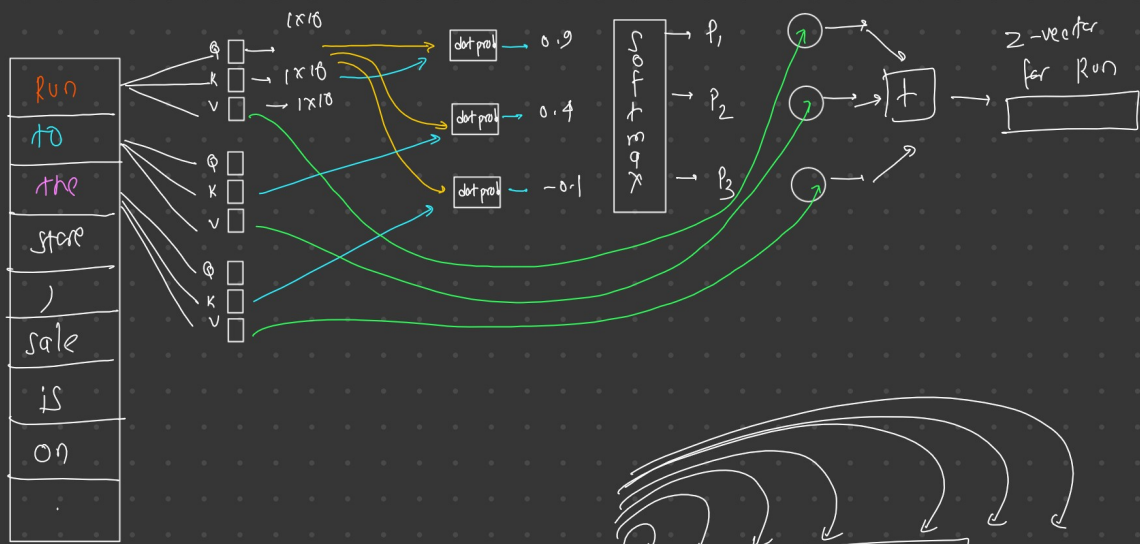
$[9 \times 10]$

Q
K
V

Q
K
V

Q
K
V





$Q \rightarrow$ information

$K \rightarrow$ information

$V \rightarrow$ information

$Q' \& K_{1-n}$

$\downarrow$
similarity w.r.t Q'

$\downarrow$
softmax

$\downarrow$
weightage of those keys w.r.t. Q

softmax $\downarrow$
scalm
 $\downarrow$
softmax $\rightarrow V$

